

A Group-Stratified Reliability Framework for Trustworthy Early-GPA Prediction in Higher Education

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The Problem: Why Accuracy Alone Isn't Enough

- Universities increasingly use GPA prediction to guide academic advising and student intervention.

- Today's evaluation asks only “is it accurate?” — hiding a real inequity: a model can look fine overall while failing the students who need help the most.

- **Three questions accuracy alone can't answer:**
  - When does a prediction become stable enough to act on?
  - Is reliability of predictions equal across performance groups?
  - Do early predictions systematically over/under-predict at-risk students?

Research Question & the Gap in the Literature

RQ: What is the earliest semester a student's final GPA can be forecasted reliably, accounting for prediction stability and error asymmetry among student groups?

Source	What It Shows	Critical Gap
212-study systematic review (Gunasekara & Saarela, 2025)	Explainability-focused EDM literature	Only 3.3% report stability; 5.6% report fairness metrics
GPA prediction survey (Alturki et al., 2020)	75% frame outcomes as pass/fail or degree class	Continuous error magnitude & group-level bias rarely quantified
Comparative GPA model study (Putpuek et al., 2018)	Very high recall for good students	Near-zero recall for poor students — error asymmetry ignored

Data used in the research

- Source: Faculty of Business Science, University of Moratuwa
- Three cohorts, 2017–2019: 271 total students
  - 178 (2017–2018) → training set
  - 93 (2019) → held-out external test set
- Outcome: Final GPA

Features
Pre-academic Z-score, English marks, A-level medium
Demographics Gender, district
Academic Semester GPA, S1–S8, department

Group	Final GPA Threshold	n
Underperforming	≤ 2.99	46
Average	2.99 – 3.29	43
Performing	> 3.29	89

Used for stratification & evaluation only

Methodology — Design & Model Selection

- Semester checkpoints: S0 (pre-academic only) → S8, but every group's reliability point was confirmed by S6
- Ridge Regression and Random Forest models were applied
- Two-stage evaluation:
  - Stage 1 = 5-fold × 10-repeat stratified Cross Validation (50 splits)
  - Stage 2 = Test on the external 2019 cohort

Metric	S0 Ridge	S0 RF	S3 Ridge	S3 RF	S6 Ridge	S6 RF
RMSE	**0.346**	0.364	**0.181**	0.182	**0.087**	0.143
SD (RMSE)	**0.063**	0.066	**0.038**	0.051	**0.036**	0.053
|Bias|	**0.001**	0.002	**0.004**	0.011	**0.001**	0.009

Methodology — **The Three-Criterion Framework**

1. Accuracy (RMSE)

- ≤ 0.25 GPA (Performing / Average)
- ≤ 0.35 GPA (Underperforming)

0.25 ≈ half a meaningful grade band (0.5) on this 4-point scale

2. Stability (SD of RMSE)

- Must show a declining trend

Catches models that look accurate on average but are fragile to which students land in which fold.

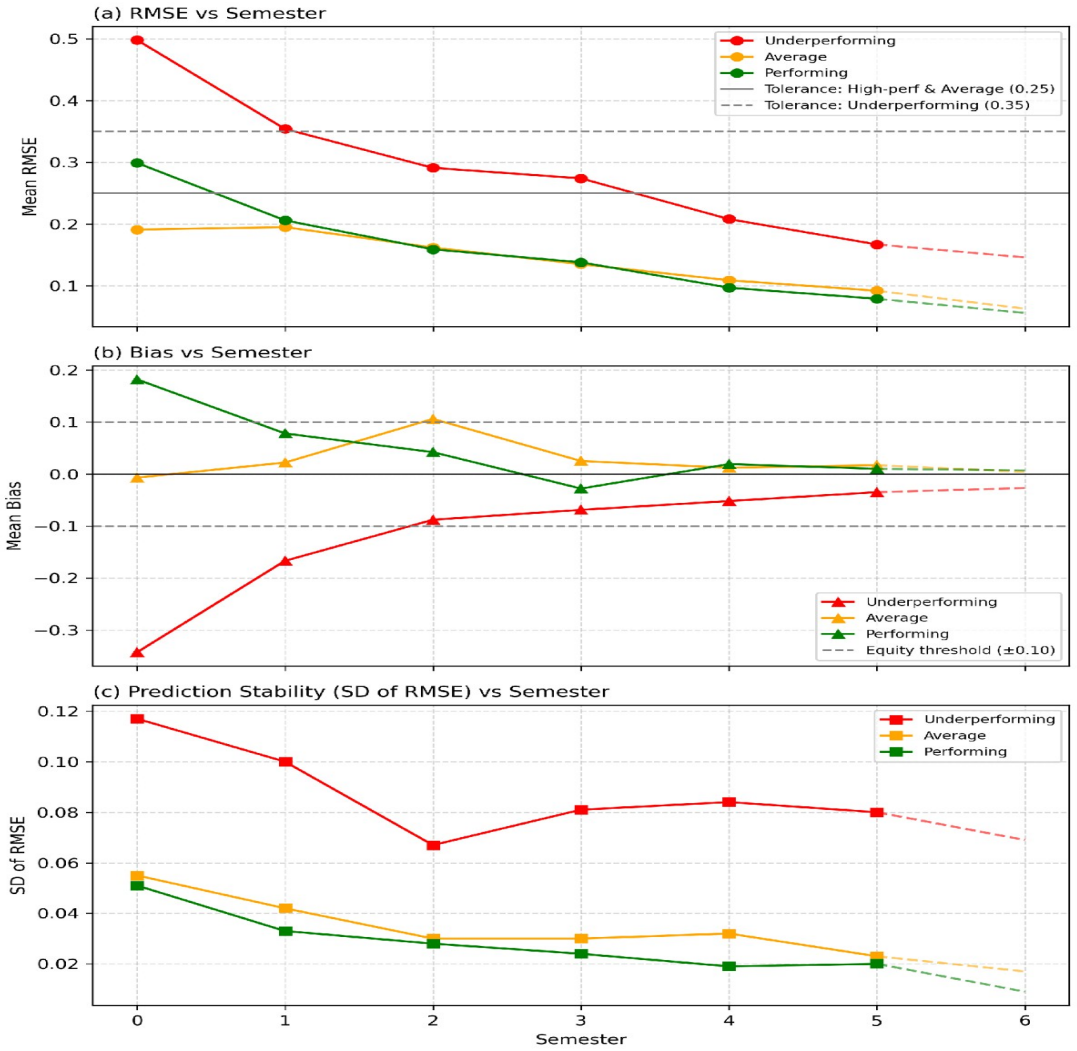
3. Equity (Group Bias)

- |mean bias| ≤ 0.10 GPA points

Caps systematic over/under-prediction to 40% of the RMSE through directional error.

These thresholds are grounded in this 4-point GPA scale. What transfers to other institutions is the three-criterion structure, not these exact numbers.

Results I — Convergence of the Three Criteria (Cross-Validation)



Performing & Average clear all three bars earliest in cross-validation.

Underperforming's SD(RMSE) stays elevated and non-monotonic through Semesters 3–4 — the binding constraint, reflecting real prediction difficulty for this smaller group (n=46).

Accuracy is not the group-defining bottleneck; stability and bias are.

Results II — External Validation (2019 Test Set)

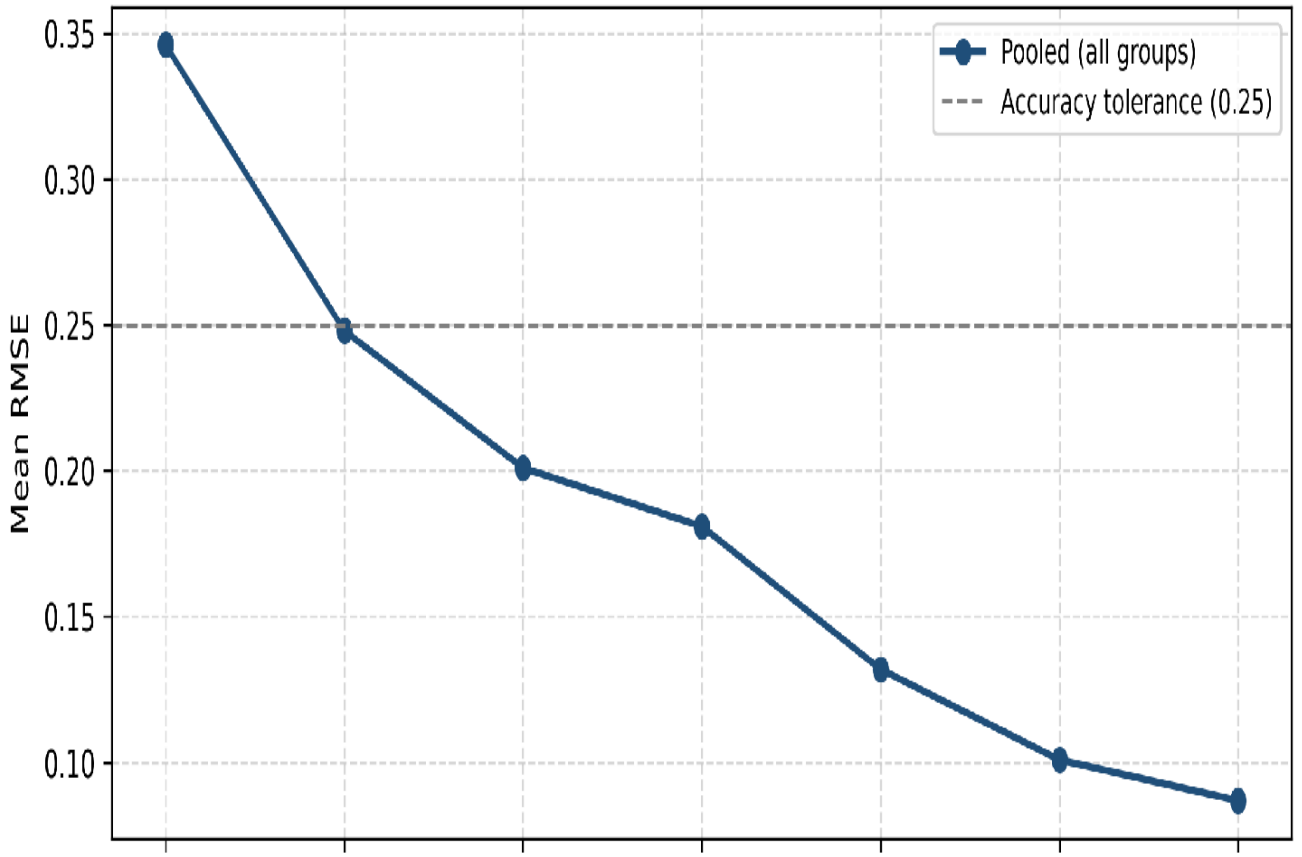
Sem	HP RMSE	HP Bias	Avg RMSE	Avg Bias	UP RMSE	UP Bias
2	0.208	-0.152	0.378	-0.321	0.603	-0.480
3	**0.142**	**-0.087**	0.228	-0.149	0.426	-0.302
4	0.091	-0.017	**0.144**	**-0.072**	0.305	-0.202
5	0.072	0.048	0.094	0.041	**0.163**	**-0.035**

HP = High-performing Avg = Average UP = Underperforming

The Earliest Reliable Semesters for the three groups are:

- High-performing – Semester 3
- Average – Semester 4
- Underperforming – Semester 5

Results — Overall Model Accuracy Measure



Pooled RMSE results look reassuring from an early stage. RMSE clears the 0.25 threshold by Semester 1-2.

Reading only this chart, you'd conclude prediction is reliable at a very early stage for everyone.

Discussion

- Accuracy alone is an insufficient deployment criterion.
- Reliable timing is group-dependent.
- **What this means for advising systems:**
  - A bias-corrected prediction (raw estimate + tier correction) can be generated automatically at early stages in the advising systems.
  - For Underperforming-tier students before Semester 5, this bias-corrected estimate, flagged as less-than-full-confidence, can still be built into advising systems as an early support signal.

Limitations & Future Directions

- **Contribution:** a reusable, equity-aware three-criterion framework.

- **Limitations:**
  - small sample, especially from 177 overall only 46 Underperforming students, ~9 per CV fold, likely driving that group's noisy stability trend.

  - Findings from this study need to be validated on a larger sample.

- **Future direction:** adds module-level grades + SHAP to find early explanatory warning signals (Sem 0–2); independent of the numeric prediction's own reliability.

Thank you.